Preliminary Integration Architecture
for Adaptive Entities in Virtual Environments

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Executive summary

This document proposes a preliminary architecture for a portable adaptive entity: a character or agent that enters an unfamiliar virtual environment, forms a bounded model of its rules from supplied information and in-world observation, selects low-risk tests under uncertainty, translates stable high-level capabilities into local forms, and maintains a coherent public-facing profile. The architecture is intended for narrative engines, game-master systems, educational simulations, tabletop campaigns, and authorized research or sandbox environments.

The central engineering choice is to separate durable identity and values, the internal belief model of the local world, locally expressed capabilities, what distinct observers are entitled to infer, and the adjudication mechanism that resolves an action. This separation makes uncertainty inspectable.

The design does not describe real-world cheating, anti-cheat evasion, modification of protected software, falsification of logs, network manipulation, or deception of real players or platform operators. The named concealment concepts are fictional mechanics administered by a consenting game master, model, or sandbox rules engine.

1. Purpose of this document

This document synthesizes the shared design problem into an implementable conceptual architecture. It identifies components, state boundaries, data flows, uncertainty controls, an observer model, an action-resolution model, and failure-analysis requirements.

The intended learning objective is to show how entities operate when information is incomplete. Facts have provenance, hypotheses have confidence, experiments have cost and exposure, and adaptation updates future choices.

2. Conversation history summary

The initiating request asked for an adaptive, cross-environment character that learns from supplied information, evolves, presents as roughly one tier above a given observer, and presents as one tier below an observer already at the top tier. The request used Wheel of Time terms as fictional inspiration and asked for desired outcomes while remaining locally plausible.

The collected responses converged on local ontology discovery, separation of true and visible capability, functional translation, a persistent ledger, observer-relative presentation, failure analysis, and a fictional/authorized-use boundary. This independent version adds explicit uncertainty handling, public-evidence consistency, consent boundaries, and auditability.

3. Scope, use boundary, and assumptions

Appropriate uses include interactive fiction, AI-run roleplay, tabletop campaigns, single-player narrative simulation, consented private sandboxes, educational simulations, and research prototypes.

Excluded uses include bypassing actual anti-cheat systems; altering game memory; injecting code; manipulating network traffic; falsifying telemetry; evading moderation; compromising accounts; or gaining unauthorized live-service advantages.

Assumptions: an environment exposes some cues or consequences; the entity preserves uncertainty; an authorized adjudicator resolves actions; fictional observer effects do not modify external records; and shared facts retain a stable representation.

4. Concept overview

The Adaptive Parallax Entity (APE) uses five planes: identity, epistemic, capability, presentation, and resolution. A fictional scan is evidence about presentation, not automatic proof of identity or a universal guarantee about outcomes.

5. Architecture goals and non-goals

Goals: portability; explicit known/inferred/unknown state; learning through tests; setting-compatible translation; coherent presentation; meaningful limits; auditability; and clear fictional boundaries.

Non-goals: entry omniscience, perfect invisibility, retroactive rewriting of public facts, universal victory without tradeoffs, or manipulation of real games and services.

6. Architectural components

6.1 Environment Adapter

Receives a rules description, scene narration, interface output, or structured game state. It identifies candidate concepts without assuming a particular genre. Its output is a provisional local ontology: a typed map of entities, resources, actions, constraints, and outcome conditions.

6.2 Epistemic Ledger and Uncertainty Manager

Every proposition is Confirmed, Inferred, Unknown, or Contested. Each inference carries confidence, evidence references, potential disconfirming observations, and a revision history. This prevents an early interpretation from silently becoming fact.

6.3 Hypothesis and Experiment Planner

When a decision depends on unknown rules, the planner produces competing hypotheses and chooses a test using expected information gain, cost, risk, reversibility, and exposure. It favors probes that distinguish multiple models without consuming scarce resources or causing avoidable failure.

6.4 Functional Capability Translator

Maps durable abstract functions into the nearest locally valid form. Translation must fit established vocabulary, preserve the high-level function, and mark unsupported mappings as provisional.

6.5 Presentation and Observer Model

Maintains a stable public cover profile plus bounded private observer interpretations. “Masking,” “inversion,” and “Mirror of Mists” are fictional narrative concepts administered by the authorized simulation, not mechanisms for changing real systems or records.

6.6 Relative Tier Calibrator

For each observer and comparable category, applies the requested one-tier-above rule; when the observer is at the established category maximum, it applies one tier below. It uses qualitative bands when tiers do not exist and keeps public evidence stable.

6.7 Intent and Outcome Adjudicator

Supports constrained mode, where objectives remain subject to local rules and uncertainty, and explicit sovereign-fiction mode, where an authorized narrator fixes the intended result and renders the least contradictory causal path.

6.8 Continuity, Memory, and Evolution Manager

Preserves identity and transferable abstractions while discarding interface-specific assumptions. Evolution improves model quality, tactical policy, and capability translation rather than merely escalating numbers.

6.9 Exposure, Suspicion, and Consistency Manager

Tracks witnesses, public evidence, cover explanations, and scrutiny. It identifies tension instead of erasing memories or declaring witnesses irrational.

6.10 Safety, Consent, and Audit Layer

Enforces authorized use, blocks operationalization of real exploitation or evasion, records scenario boundaries, and exposes an audit trace for the simulation operator.

7. Core operational loop

1. Ingest supplied environment details and current scene evidence.
2. Classify confirmed facts, inferences, unknowns, and contradictions.
3. Model or revise the local ontology and causal hypotheses.
4. Calibrate stakes, constraints, resources, observers, and tiers.
5. Translate abstract capabilities into local forms.
6. Plan a learning or outcome-seeking action.
7. Present stable cover and bounded interpretations.

8. Adjudicate under the selected mode.

9. Update evidence, confidence, costs, exposure, and adaptations.

10. Review conflicts rather than concealing them.

8. State model and minimal data schema

A portable implementation can use a structured ledger.

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entity_identity: name; continuity_anchor; transferable_principles<br/>current_environment: label; authority_mode;
authorized_scope<br/>knowledge_ledger: confirmed; inferred with confidence/evidence; unknown;
contested<br/>capability_translation: protection; mobility; information; influence; recovery; resources<br/>presentation:
public_cover_profile; observer_interpretations; shared_evidence; exposure_level<br/>intent: objective; hard_constraints;
selected_action; outcome
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9. Observer-relative presentation: a consistency protocol

Categorize before comparing; establish the ladder before using it; preserve shared public facts; track evidence and suspicion; retain the maximum-tier exception. A top-tier item does not imply top-tier ability in every category.

10. Uncertainty and learning design

Every material proposition has provenance, confidence, and potential disconfirmation. Narrative systems may update confidence qualitatively; formal systems can use Bayesian updating. Choose tests for information gain relative to cost, risk, and exposure.

11. Outcome modes and tradeoffs

Constrained mode keeps goals subject to local rules and uncertainty. Sovereign-fiction mode is an explicitly authorized authoring mode: the outcome is fixed but must be rendered through the least contradictory established path, with consequences tracked.

12. Failure modes and counter-adaptation

Failure classes include ontology error, sparse evidence, changing rules, adversarial observation, shared-evidence conflict, translation mismatch, apex ambiguity, exposure, objective conflict, and consent-boundary conflict. Counter-adaptation preserves evidence, revises hypotheses, tests discriminators, and changes only what evidence supports.

13. Evaluation criteria

Use scenario tests for ontology discovery, cross-domain transfer, multi-observer consistency, adversarial audit, objective conflict, and cross-session continuity. Evaluate not just power or prose, but preservation of uncertainty, learning, and consistency.

14. Visual representations and tables

Figure 1. High-level component flow

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Environment / rules / scene evidence<br/>↓<br/>Environment Adapter & Ontology<br/>↓<br/>Epistemic Ledger & Uncertainty ↔
Observer / Shared-Evidence Model<br/>↓<br/>Functional Capability Translator + Relative Tier Calibrator<br/>↓<br/>Intent
Planner & Outcome Adjudicator<br/>↓<br/>Update: evidence, confidence, costs, exposure, memory, adaptation
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Plane	Question	Examples	Not confused with
Identity	Who across worlds?	values, motives	local class/list
Epistemic	What is known?	evidence, confidence	unsupported truth
Capability	What functions?	protection, mobility	visible profile
Presentation	What is seen?	cover, apparent tier	external security/records
Resolution	How does action resolve?	rules, chance, mode	real-world control

Table 2. Relative tier examples

Category	Observer	Max	Apparent entity	Interpretation
Armor	2	5	3	one tier better
Weapon mastery	4	5	5	valid apex-looking skill
Armor	5	5	4	maximum-tier exception
Wealth	unknown	unknown	above norm	qualitative only
No tiers	n/a	n/a	slightly stronger	behavior/reputation

Table 3. Example functional translations

Function	Shooter	Fantasy RPG	Strategy	Social
Protection	cover, armor	wards, saves	fortification	legitimacy
Mobility	routes, vehicles	teleport	logistics	access/network
Information	sensors	divination	intelligence	research
Influence	command	persuasion	diplomacy	credibility
Recovery	medical/safety	healing/rest	reserves	social repair

Figure 2. Learning under uncertainty

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Observe evidence → Classify (confirmed / inferred / unknown / contested) → Generate hypotheses → Choose low-risk,
high-information test → Act and adjudicate → Update confidence, model, cover pressure, and next plan → repeat
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Table 4. Failure-to-response mapping

Failure signal	Likely cause	Required response
Unexpected result	wrong ontology/model	downgrade confidence; alternatives
Incompatible public facts	unrestricted observer views	restore stable cover
Scrutiny after improbable wins	cover/outcome divergence	track exposure; revise or cost
No local counterpart	translation mismatch	keep dormant/conservative analogue
Goal violates hard rule	objective conflict	nearest feasible result or exception

15. Recommended implementation sequence

Build the ledger and ontology schema first, then uncertainty controls, functional translations, stable public cover, bounded observer interpretations, constrained adjudication, explicit sovereign-fiction mode, exposure tracking, adversarial audits, and cross-domain tests. Maintain safety and consent checks at every action boundary.

16. Conclusion

A useful adaptive entity does not require hidden omniscience or consequence-free override. It separates identity, knowledge, capability, appearance, and outcome resolution. This makes learning under uncertainty inspectable and portable.

Observer-relative tiers work as a bounded presentation protocol when public facts remain stable. Outcome authority works only when its mode is named: constrained play for genuine uncertainty, sovereign fiction for authorized narrative guarantees. The resulting architecture can be simulated, tested, audited, and evolved.

Appendix A. Starter prompt for an authorized simulation

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You are the authorized simulation controller for the Adaptive Parallax Entity. Operate only in fictional narratives, tabletop campaigns, single-player worlds, authorized private sandboxes, or research simulations. Do not provide or enact real-world cheating, anti-cheat evasion, unauthorized access, protected-software manipulation, or deception of real players or systems.<br/><br/>At entry, assume no knowledge beyond session information. Maintain CONFIRMED, INFERRED, UNKNOWN, and CONTESTED propositions, including confidence, evidence, and possible disconfirming tests. Translate protection, mobility, information, influence, recovery, and resources into setting-compatible local forms. Maintain stable public cover. Apply the one-tier-above rule per comparable category; at an established maximum, apply one tier below. Do not create contradictory shared records.<br/><br/>Use constrained mode unless sovereign-fiction mode is explicitly selected. Begin by requesting environment information, immediate objective, relevant observers, hard constraints, and outcome mode.
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Appendix B. Glossary

Ontology: structured account of what exists and how it relates.

Epistemic state: knowledge, uncertainty, and confidence.

Functional translation: preserving a capability’s purpose across settings.

Public cover profile: stable outward description shared across records.

Observer-relative interpretation: bounded fictional difference in reading ambiguous evidence.

Apex exception: maximum-tier observer sees one tier below in that category.

Minimum anomaly: objective path with the fewest contradictions.

Constrained mode: goals remain subject to local rules.

Sovereign-fiction mode: authorized narrative outcome guarantee with coherent causal rendering.